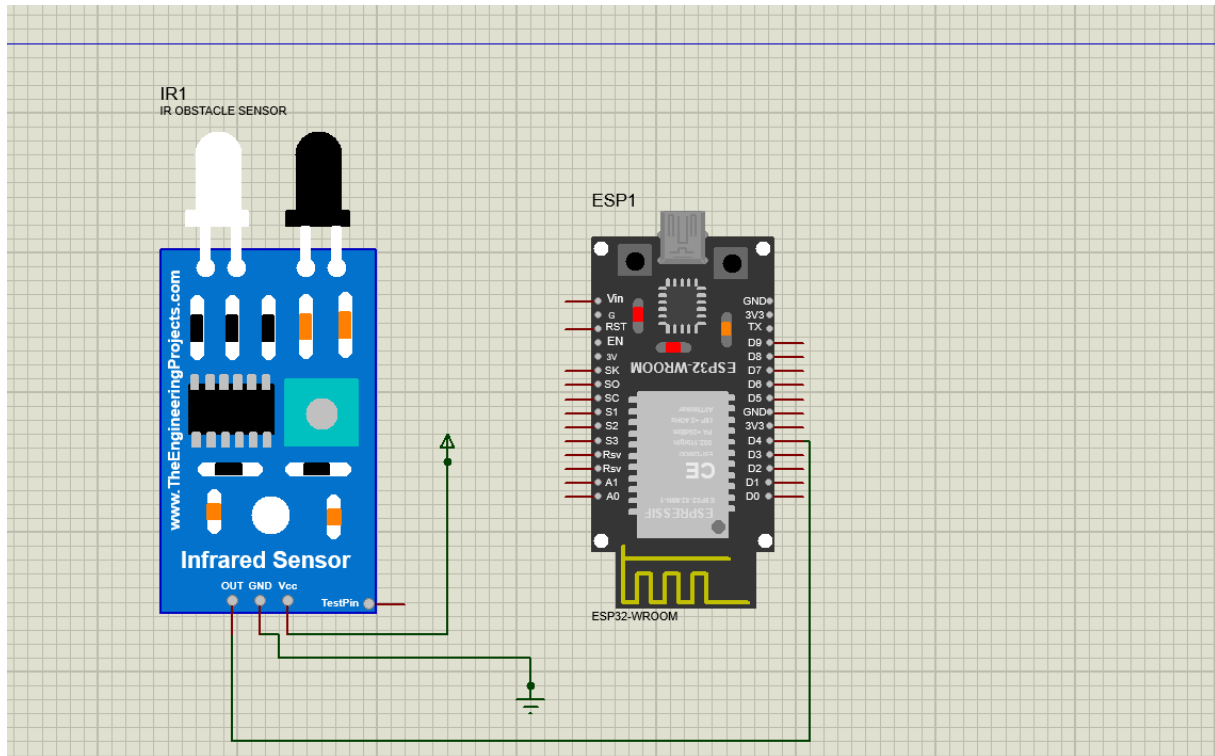


Hardware Setup



Hardware Required

- ESP32 Development Board
- IR Sensor
- Jumper Wires
- USB Cable

Software Required

- Arduino IDE
- ESP32 Board Package
- Wifi Connection
- Blynk App Smartphone

Wiring Connection

IR sensor	ESP32
Pinout	GPIO4
Vcc	3.3V

GND	GND
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Installation Setup (IR Sensor with ESP32)

Step 1: Install Arduino IDE

- Download and install the Arduino IDE from the official website.
 - Launch Arduino IDE after installation.
-

Step 2: Install ESP32 Board Package

1. Open Arduino IDE → File → Preferences.
2. In Additional Boards Manager URLs, add:

https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

3. Click OK.
 4. Go to Tools → Board → Boards Manager.
 5. Search for ESP32.
 6. Install the ESP32 by Espressif Systems package.
-

Step 3: Connect the IR Sensor to ESP32

Make the wiring connections as shown below:

IR Sensor ESP32

OUT GPIO4

VCC 3.3V

GND GND

Ensure all connections are secure before powering the ESP32.

Step 4: Upload the Program

1. Connect the ESP32 board to your PC using a USB cable.
 2. Open the IR Sensor source code in Arduino IDE.
 3. Select Tools → Board → ESP32 Dev Module.
 4. Select the correct COM Port under Tools → Port.
 5. Click the Upload button and wait for the upload to complete.
-

Step 5: Configure Wi-Fi and Blynk

1. Open the source code.
 2. Enter your:
 - Wi-Fi SSID
 - Wi-Fi Password
 - Blynk Authentication Token
 3. Save the changes.
 4. Upload the updated code to the ESP32.
-

Step 6: Run the System

1. Open Serial Monitor.
2. Set the baud rate to 115200.
3. Place an object in front of the IR sensor.
4. The sensor status will be displayed in the Serial Monitor.
5. Open the Blynk App to monitor the object detection status in real time.

Expected Output:

- Object detected → Status updates in Blynk Dashboard.
- No object detected → Status changes accordingly.
- Real-time monitoring through Wi-Fi connection.